

Swastik Majumder

Curriculum Vitae

The Ohio State University

Department of Physics & Center for Cosmology and AstroParticle Physics (CCAPP)

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Education

Doctor of Philosophy in Physics

2025–Present

The Ohio State University, Columbus, Ohio, United States.

Bachelor of Science and Master of Science in Physics

2020–2025

Indian Institute of Science Education and Research Kolkata, India.

Research Interests

In-medium quantum field theory: Perturbative quantum chromodynamics at finite temperature and density through effective field theories.

Statistical mechanics: Equilibrium and nonequilibrium behavior of classical many body systems.

Publications

4. **S. Majumder** and T. Gorda.
Hot and dense QED at $N^3\text{LO}$: infrared sector.
In preparation (2026).
3. **S. Majumder** and M. Barma.
Stochastic systems with Bose-Hubbard interactions: Effects of bias on particles on a random comb.
Physical Review E **113**, 044137 (2026).
[Journal](#) | [arXiv:2601.09962](#).
2. **S. Majumder** and M. Barma.
Stochastic systems with Bose-Hubbard interactions: Effects of bias on particles on a one-dimensional lattice.
Physical Review E **113**, 044136 (2026).
[Journal](#) | [arXiv:2508.03344](#).
1. **S. Majumder** and N. Sasakura.
Three cases of complex eigenvalue/vector distributions of symmetric order-three random tensors.
Progress of Theoretical and Experimental Physics **2024**, 093A01 (2024).
[Journal](#) | [arXiv:2408.01030](#).

Presentations

Poster presentation

October 2026

S. Majumder and T. Gorda.

Thermal corrections to the equation of state of dense QED at high perturbative orders

American Physical Society Division of Nuclear Physics Fall Meeting, Philadelphia, Pennsylvania.

Poster presentation

August 2026

S. Majumder and T. Gorda.

The $N^3\text{LO}$ Mixed Contribution to the Pressure of a Hot and Dense Relativistic QED Plasma

Department of Physics Poster Session, The Ohio State University, Columbus, Ohio.

Research Experience

The Ohio State University <i>Graduate Research Associate</i> , research supervisor: Dr. Tyler Gorda.	May 2026–Present
Tata Institute of Fundamental Research, Hyderabad <i>Visiting Research Student</i> , research supervisor: Prof. Mustansir Barma.	May 2024–June 2025
Yukawa Institute for Theoretical Physics, Kyoto University <i>Visiting Research Student</i> , research supervisor: Dr. Naoki Sasakura.	May 2023–May 2024

Teaching Experience

The Ohio State University <i>Graduate Teaching Associate</i>	
Electricity and Magnetism, Waves, Optics, and Modern Physics.	Spring 2026
Electricity and Magnetism, Waves, Optics, and Modern Physics.	Autumn 2025
Indian Institute of Science Education and Research Kolkata <i>Teaching Assistant</i>	
Statistical Mechanics	Spring 2025
Probability and Statistics II	Spring 2025
Condensed Matter Physics	Autumn 2024
Statistical Mechanics	Spring 2024

Academic Service

President, Physics Graduate Student Council The Ohio State University	2025–2026
Leadership team, Science Club Indian Institute of Science Education and Research Kolkata	2021–2022

Awards and Honors

Graduate Associate Teaching Award nomination The Ohio State University.	2025
Silver Medal , University Physics Competition	2022
Finalist , International Astronomy and Astrophysics Competition	2021

References

Dr. Tyler Gorda Department of Physics and Center for Cosmology and AstroParticle Physics (CCAPP), The Ohio State University, United States.
Prof. Mustansir Barma Tata Institute of Fundamental Research, Hyderabad, India.
Dr. Naoki Sasakura Yukawa Institute for Theoretical Physics, Kyoto University, Japan.

Prof. Arindam Kundagrami

Department of Physical Sciences,

Indian Institute of Science Education and Research Kolkata, India.

Software and Programming

Python, Mathematica, L^AT_EX.

Languages

Bengali: Mother tongue.

Hindi and English: bilingual proficiency.